

# Outcome

Bounded null-sequence guidance improves the metric we ultimately care about: oracle best-of-100 contact accuracy.

On 81 eval-val proteins untouched during method selection, all-token poly-Ala guidance at  $\gamma=0.5$  raises validity-gated oracle R-precision from 0.5331 to 0.5554 all-range and from 0.5279 to 0.5474 long-range. Paired bootstrap intervals exclude zero.

The literal native/null probability ratio does not work: it falls to 0.4774/0.4713 and becomes unreliable. The useful decoder retains native logits and adds a bounded contrast.

# Method and amended endpoint

For each rollout, native and same-length poly-Ala prompts share the exact emitted suffix. Only the native stream samples:

$\text{guided} = \text{native} + 0.5 * (\text{native} - \text{polyAla})$

Development used 16 natural proteins and 15 fold-switch pairs. The full comparison used the other 81 eval-val proteins; no eval-test data was read.

After the original full arms launched, the owner clarified that consensus accuracy was secondary and oracle best-of-100 was primary. We froze that amendment before adding pure-ratio and T=1.1 controls. An unfinished rollout or any rollout with a malformed contact statement scores zero.

# Primary result

At N=100 on the untouched 81 proteins:

Bounded guidance: 0.5554 all / 0.5474 long.

Ordinary iid T=1.0: 0.5331 / 0.5279. Overlap-matched T=0.8: 0.5384 / 0.5343. High-diversity T=1.1: 0.5169 / 0.5112. Pure native/null ratio: 0.4774 / 0.4713.

Guidance minus iid is +0.02235 all-range (95% CI +0.01588 to +0.02899) and +0.01952 long-range (+0.01119 to +0.02800).

The gamma=0 path reproduces the established exp277 score across all 97 eval-val proteins within 0.0007, validating the implementation.

# Why it works

The oracle gain comes from stronger individual samples, not a broader pool.

Guidance raises mean valid single-rollout R-precision from 0.4073 to 0.4340, while pairwise Jaccard rises from 0.2846 to 0.3334 and true-contact union recall falls from 0.9364 to 0.9175.

Pure ratio does create breadth—union recall 0.9612—but over-generates: 249/8,100 invalid rollouts, 558 malformed statements, and 353 contacts per rollout. Its best map is much worse.

So the background stream is a useful contrastive quality signal only while the native distribution remains the base.

# Compute and implications

One hundred guided rollouts fit the same H100 time as a median 132 iid rollouts. Guidance still wins that equal-time comparison by +0.01714 all / +0.01415 long, with both intervals excluding zero. It even beats 200 iid rollouts all-range; the long-range 200-rollout interval includes zero.

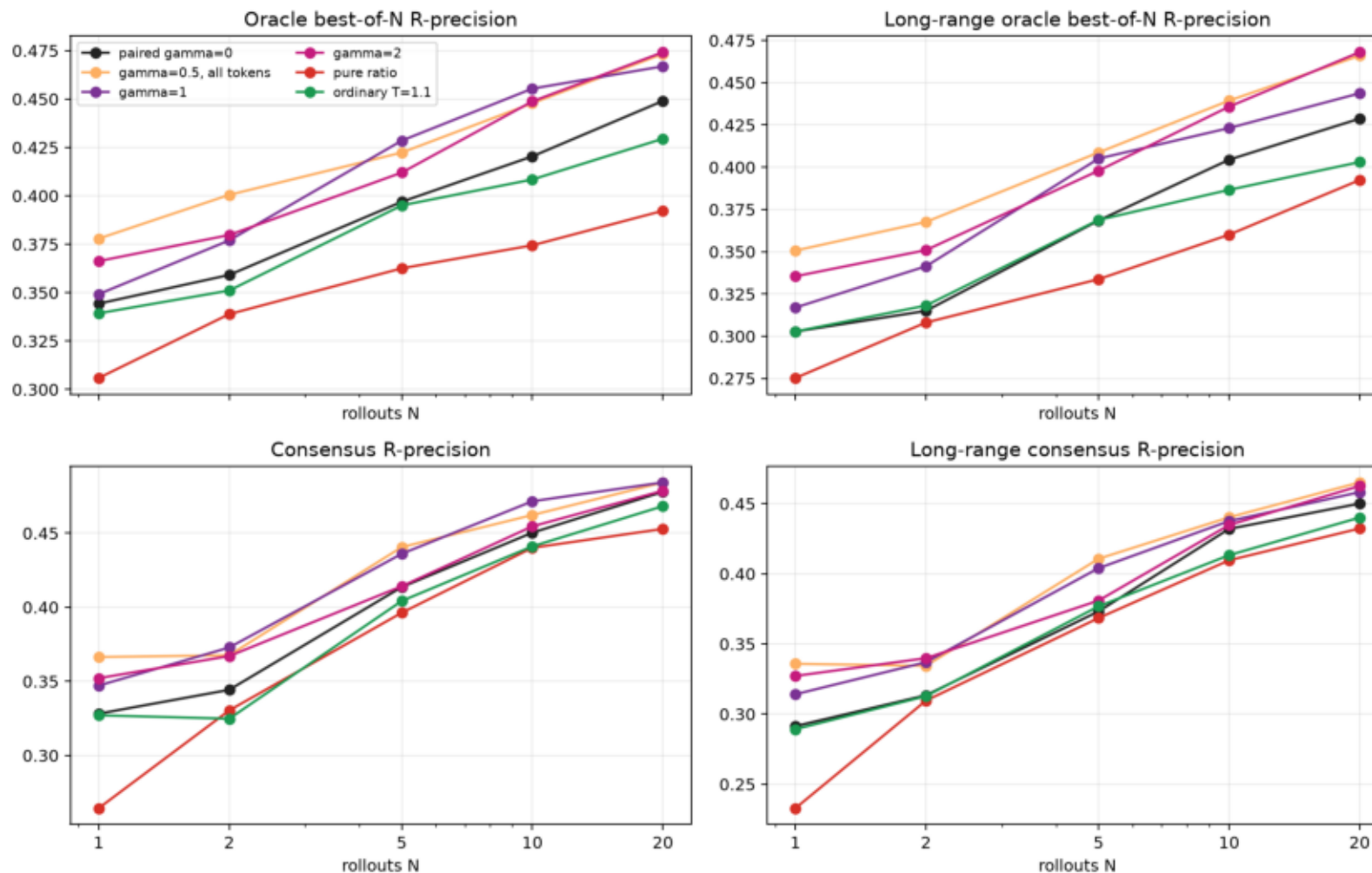
Fold switching is secondary: guidance scores 6/29 oracle and 3/29 blind versus iid 6/29 and 2/29. Pure ratio reaches 7/29 and 4/29 but emits 5,329 malformed statements, so this is not a usable win.

Conclusion: keep bounded  $\gamma=0.5$  guidance for oracle-oriented sampling, reject pure-ratio decoding, and next build a truth-free selector that can realize the improved upper tail.

# 01\_dev\_curves.png

Mean over 16 length-stratified eval-val development proteins. The top row is validity-gated oracle accuracy; moderate guidance improves it, while pure ratio and higher temperature do not. Consensus accuracy is shown as secondary context.

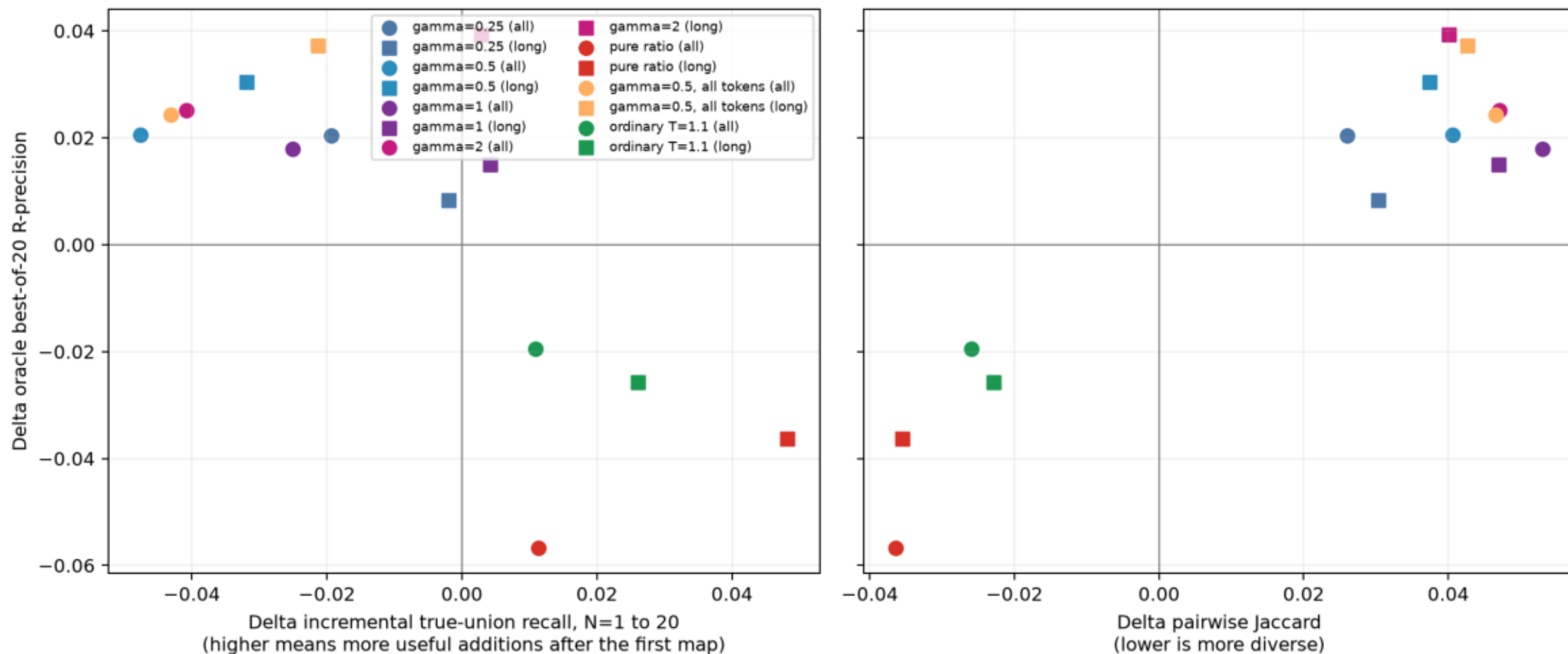
Development screen: moderate guidance improves oracle accuracy



## 02\_accuracy\_diversity\_tradeoff.png

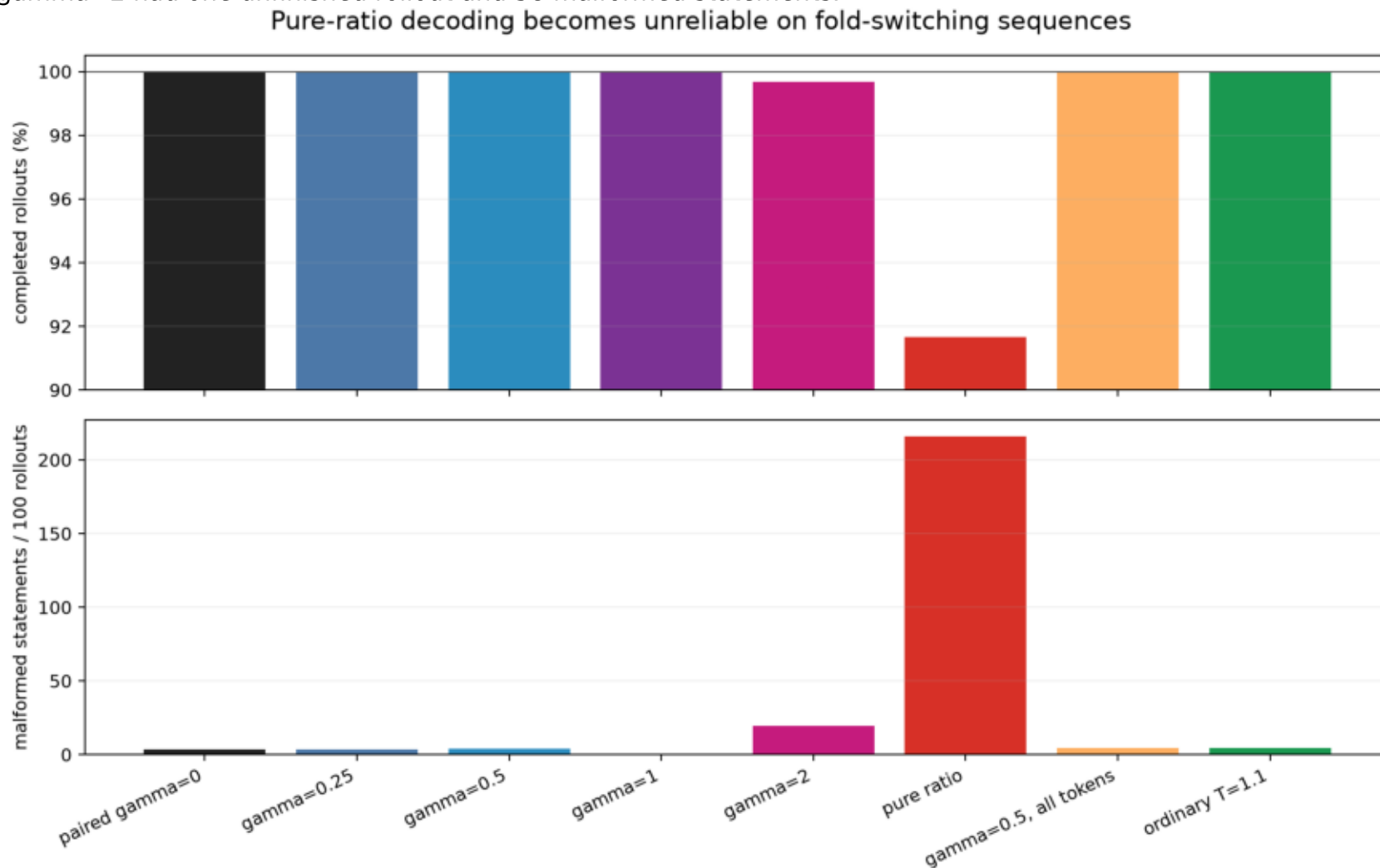
Paired mean deltas versus gamma=0 on the same 16 proteins. Incremental coverage subtracts each arm's N=1 coverage from its N=20 coverage. Circles are all-range and squares long-range. Moderate guidance raises oracle accuracy despite higher overlap and no gain in incremental union coverage.

Oracle gains come from stronger samples, not a broader contact union



## 03\_foldswitch\_reliability.png

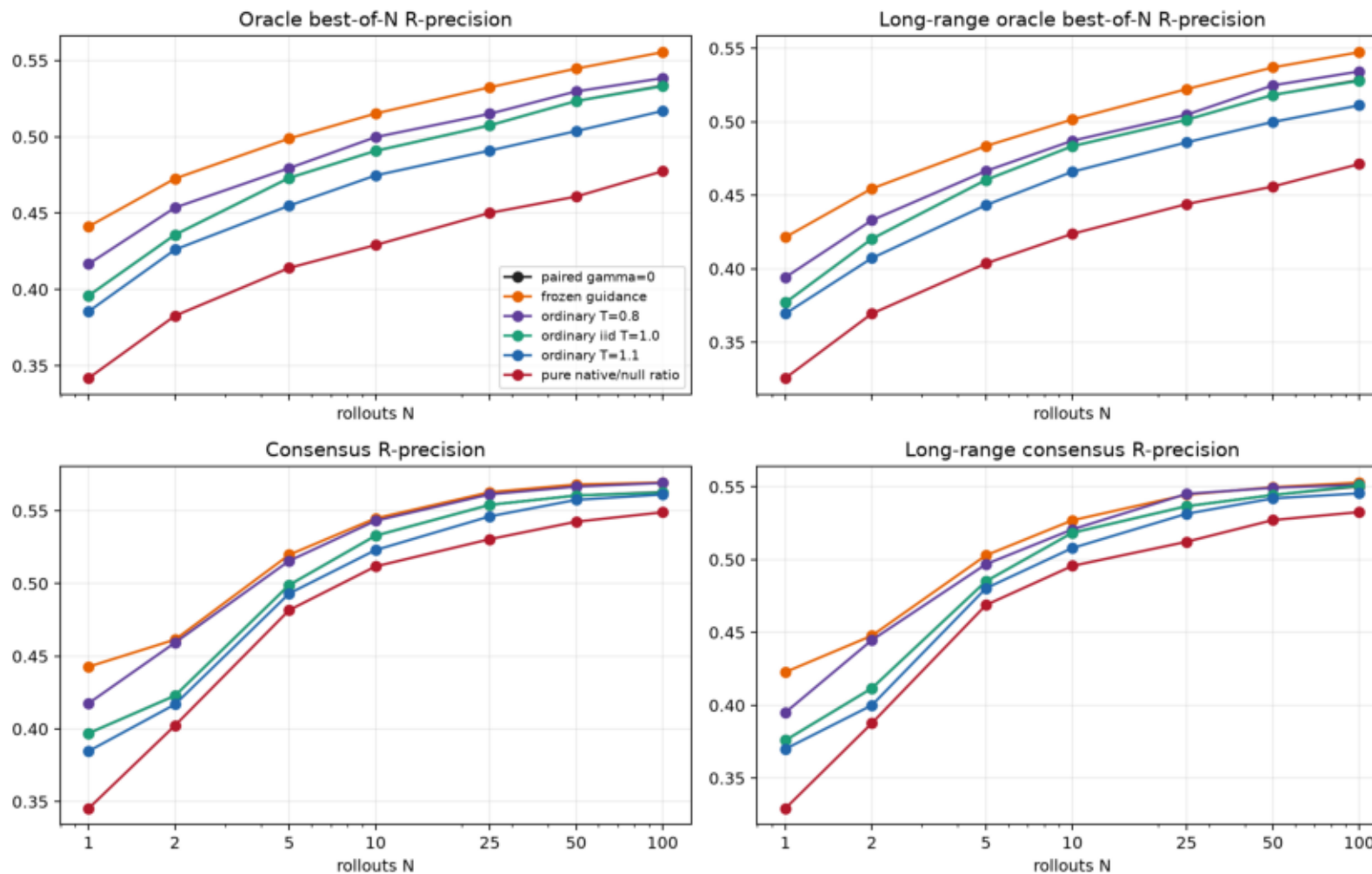
Fifteen primary fold-switch development pairs, 20 rollouts each. Pure ratio had 25/300 unfinished rollouts and 648 malformed contact statements; gamma=2 had one unfinished rollout and 58 malformed statements.



## 04\_heldout\_curves.png

Mean curves on the 81 non-development eval-val proteins. The top row is the primary endpoint: the best valid rollout in each prefix, with unfinished or malformed rollouts assigned zero. Ordinary iid and T=1.1 use their first 100 of 200 rollouts.

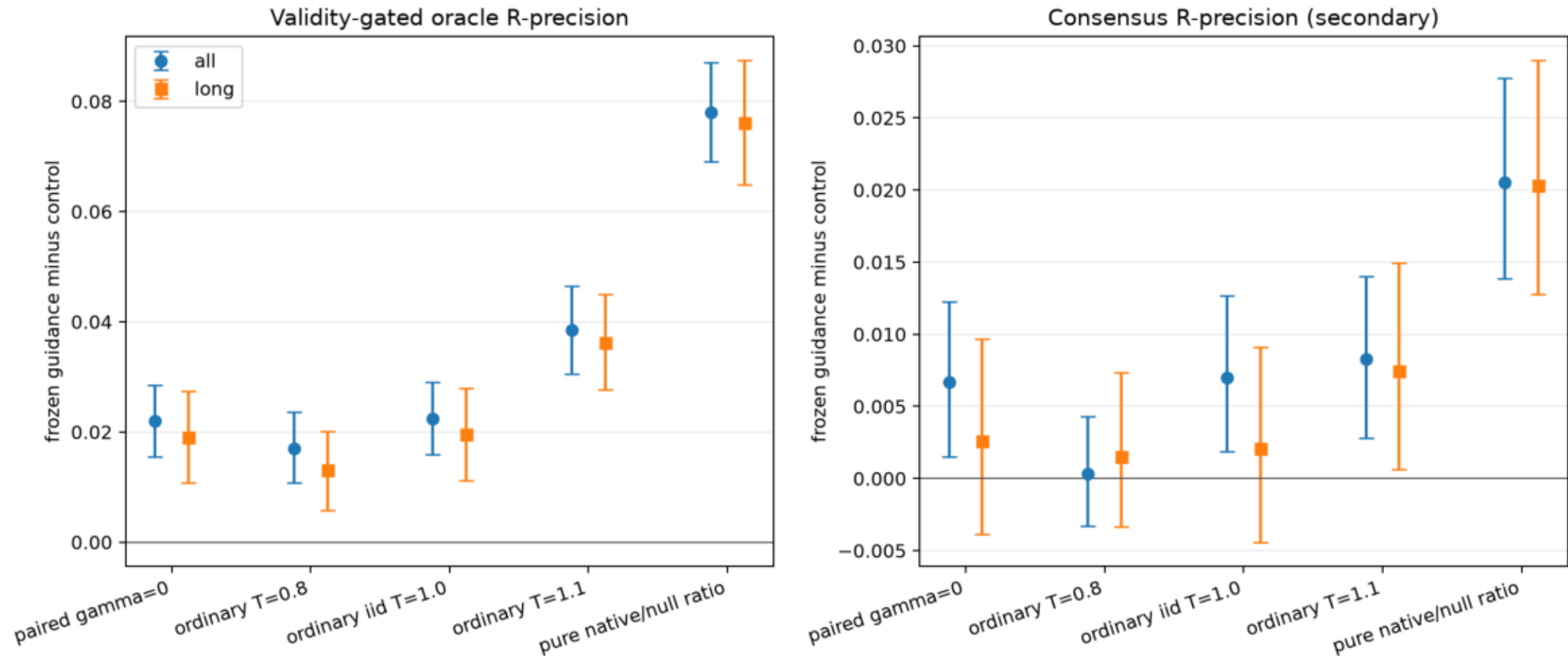
Oracle-first evaluation on 81 untouched eval-val proteins



## 05\_heldout\_primary\_deltas.png

Frozen-guidance deltas on 81 proteins. Oracle best-of-100 is primary; consensus accuracy is retained only as context. Error bars are paired protein-bootstrap 95% intervals.

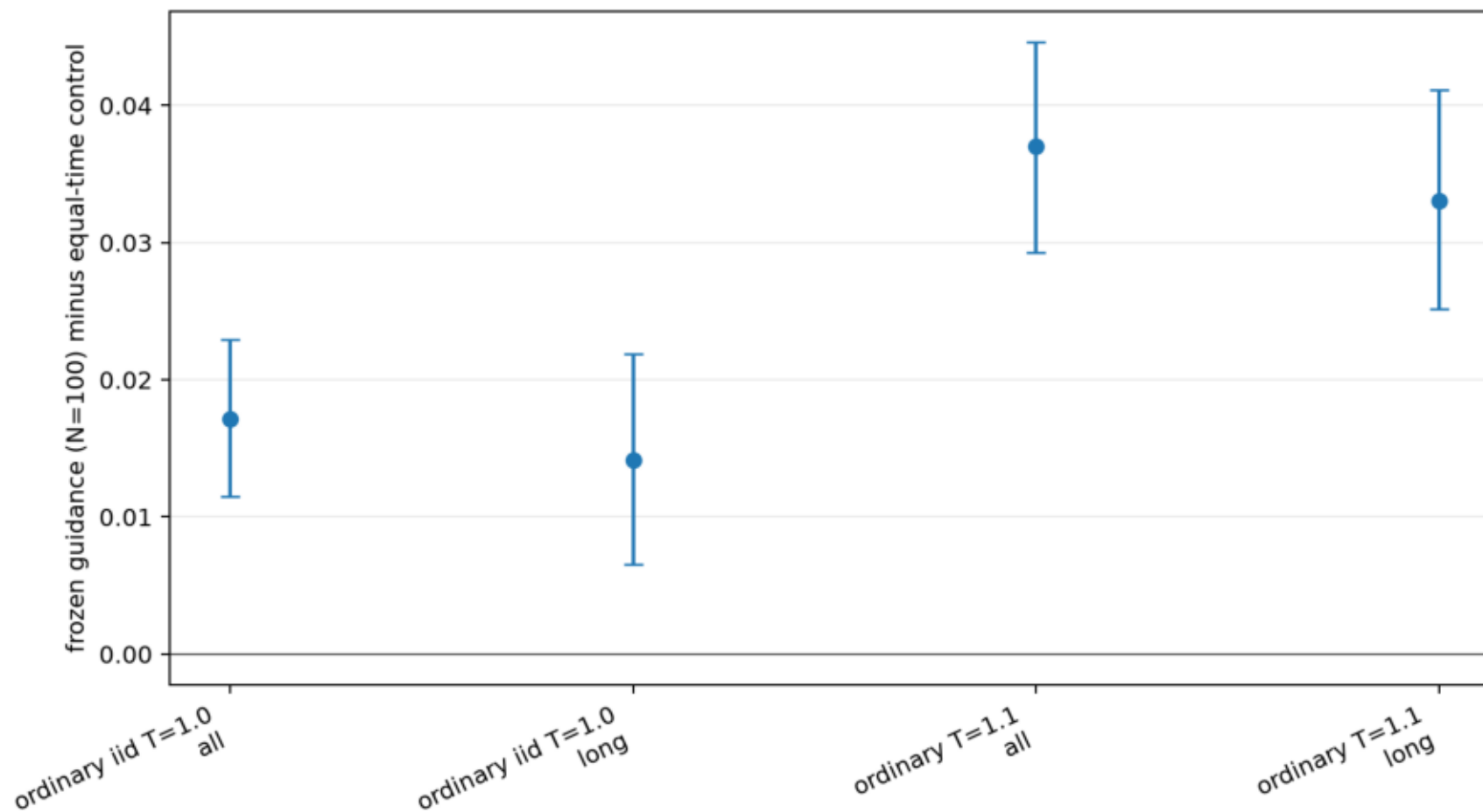
Held-out paired effects with 95% protein-bootstrap intervals



## 06\_equal\_time.png

Each protein compares 100 paired guided rollouts with the number of ordinary  $T=1.0$  or  $T=1.1$  rollouts fitting the measured guided inference time, interpolated from each 200-rollout timing. The endpoint is validity-gated oracle R-precision.

Equal-H100-time comparison using measured per-protein throughput



## 07\_foldswitch\_modes.png

Dual-mode coverage on the fixed 29-pair fold-switching stress set. This cohort is not a pristine benchmark; oracle and reference-blind top-16 definitions match exp304.

